MacroAlpha

Global Macroeconomic Sensitivity & Corporate Performance Analysis System

Database Systems Design Project Report
Course: ST207

Candidate Numbers: **66469, 66503, 73702**
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Abstract

This report presents the implementation of MacroAlpha, a database application for global macroeconomic sensitivity and corporate performance analysis. The system integrates financial and macroeconomic data extracted from Bloomberg Terminal (using Bloomberg Spreadsheet Builder and Terminal functions) into a normalized relational database covering 1,274 companies across 5 countries over a 20-year horizon (2005–2024). The implementation includes 15 SQL queries across 5 use cases covering market concentration analysis, corporate leverage cycles, rate-shock stress testing, macro lead-lag relationships, and sector rotation analysis. Key empirical findings align with established academic research: counter-cyclical deleveraging patterns consistent with credit cycle theories [2, 3], market concentration dynamics during crises matching volatility research [11, 12], sector performance variations across inflation regimes supporting inflation hedging literature [6], and revenue volatility patterns confirming idiosyncratic risk studies [8]. The project demonstrates database design, SQL development, Python data engineering, and query optimization techniques, while contributing empirical validation of theoretical frameworks using real-world financial data.

Keywords: Database Design, Financial Data Analysis, SQL Queries, Data Pipeline, Performance Optimization, Empirical Finance, Credit Cycles, Market Concentration

1 Introduction

1.1 Project Goal

MacroAlpha seeks to quantify macroeconomic sensitivity across multiple layers of aggregation—ranging from broad sectors to individual firms—as economic conditions evolve through various cycles. By identifying defensive versus cyclical exposures, measuring inflation resilience, and assessing rate-shock solvency risk, the system provides a robust framework for systematic financial analysis. Crucially, MacroAlpha extends beyond purely descriptive insights to support "what-if" scenario analysis. Through reproducible SQL queries operating on real-world datasets, researchers can explore counterfactual outcomes under varying economic assumptions, thereby validating and extending established theoretical frameworks in empirical finance by applying them to comprehensive real-world financial data spanning multiple countries and economic cycles.

1.2 Technical Complexity

The project addresses several technical challenges inherent in financial data analysis. First, it handles multi-granularity time series alignment, integrating weekly price data with quarterly macroeconomic indicators and annual financial statements, requiring careful temporal matching strategies. The implementation leverages advanced SQL techniques, including window functions such as LAG, LEAD, ROW_NUMBER, NTILE, and manual statistical calculations (variance, covariance), alongside complex joins and derived metrics computed on-the-fly during query execution.

A critical design principle is maintaining point-in-time correctness throughout the database. Index membership uses `as_of_date` to track composition changes, ensuring only companies actively trading at each date are analyzed. For financial statement data, the analysis uses `period_end_date` with an implicit assumption of reporting lags: quarterly financials are assumed available 45 days post-period-end, annual financials 60–90 days post-period-end. This standardized lag assumption ensures global consistency across diverse regulatory jurisdictions, effectively eliminating look-ahead bias while maintaining robust cross-market comparability. Macro indicators (GDP, CPI, policy rates) are treated as available in the month of publication.

For return calculations, the analysis uses **arithmetic approximations** for periodization: weekly returns are converted to monthly estimates using $\times$ 4.33 weeks/month and annualized using $\times$ 52 weeks/year. This approach provides significant computational efficiency in SQL contexts compared to geometric compounding, while maintaining the directional accuracy required for regime classification and cross-sectional ranking.

Finally, the system enables scenario-based stress testing, allowing users to simulate the impact of rate shocks and different inflation regimes on corporate performance and market dynamics.

1.3 Data Coverage

Table 1: Database Statistics

Table	Records	Time Range
companies	1,274	–
index_membership	12,069	2005–2024
prices_weekly	1,212,285	2005–2024
financials	90,911	2005–2024
macro_indicators	109,652	2005–2024
interest_rates	9,941	2005–2024

2 Data Sources & Acquisition

2.1 Bloomberg Terminal Data

All data was extracted from the Bloomberg Terminal using a combination of the Spreadsheet Builder for bulk time-series exports in Excel format and Terminal functions that provide direct query access to specific data types:

1. **Equity Prices**: Bloomberg Spreadsheet Builder → Time Series Table. Weekly frequency (Friday close) fields: `PX_LAST`, `DAY_TO_DAY_TOT_RETURN_GROSS_DVDS`.
2. **Financial Statements**: `<FA>` fundamental data fields for standardized income statement, balance sheet, and cash flow items (annual and quarterly).
3. **Macro Indicators**: `<ECST>` economic statistics for GDP, CPI, and policy rates across the five target countries (US, UK, DE, JP, CN).
4. **Index Membership**: `MEMB <GO>` function for historical FTSE 100 constituents with point-in-time effective weights and shares.

2.2 Data Files

Table 2: Source Data Files

File	Description
company_master.csv	Company metadata (ticker, name, GICS, country)
index_membership_snapshot.csv	Historical index constituents with weights
price_weekly.xlsx	Weekly closing prices and returns
financials_annual.xlsx	Annual financial statements (7 fields)
financials_quarterly.xlsx	Quarterly financial statements
*_macros_2024∼2005.xlsx	Macro indicators for 5 countries
5 countries 10y yield...xlsx	10Y yields and policy rates

2.3 Rationale for Weekly Frequency

Weekly data provides an optimal balance for this analysis framework. The frequency aligns naturally with quarterly macroeconomic indicators and annual or quarterly financial statements, enabling straightforward temporal matching without excessive interpolation. Weekly aggregation effectively filters intraday volatility and market microstructure noise that can obscure longer-term patterns, while preserving sufficient granularity for all proposed analyses including correlations and rolling window calculations. From a practical standpoint, weekly frequency reduces the dimensionality of the dataset—with over 1,000 tickers spanning 20 years, daily data would create an unwieldy dataset without providing meaningful additional insights for the macroeconomic sensitivity analyses conducted in this project.

3 Database Schema Design

3.1 Entity-Relationship Model

The database architecture is partitioned into four primary logical domains: the **Company Domain** tracks entity metadata and static identifiers; the **Market Data Domain** encompasses weekly performance and granular financial statements; the **Macro Domain** provides the broader economic context through indicators and rates; and the **Index Domain** manages point-in-time membership and constituent weighting.

3.2 Table Definitions and Business Logic

Companies Table Stores static metadata for all companies. The `is_active` flag manages the lifecycle of delisted companies.

Index Membership (Point-in-Time) Captures historical index constituents. The composite primary key `(index_id, company_id, as_of_date)` ensures that membership changes over time are accurately tracked, allowing for precise reconstruction of any historical portfolio. This handles the "survivorship bias" problem by maintaining records of companies that eventually left the index.

Prices_Weekly Table Stores weekly historical pricing data. Total return fields (PX_LAST, DAY_TO_DAY_TOT_RETURN) enable capture of market dynamics while filtering high-frequency noise.

Financials Table Stores raw financial statement items extracted from Bloomberg Fundamental data (`<FA>`). Financial ratios (ICR, Debt-to-Revenue) are not stored but computed on-the-fly to maintain data integrity and consistent definitions.

Macro_Indicators Table Stores quarterly/monthly macro data (GDP, CPI) for 5 countries. The `value_type` field distinguishes between nominal levels, YoY growth, and QoQ growth.

Interest_Rates Table Tracks national policy rates and 10-year sovereign yields, providing the benchmark for hurdle rate and solvency stress-testing.

3.3 Analytical Methodology: Financial Sector Exclusion

Several use cases explicitly exclude companies in the **Financials** sector from EBITDA-based analyses. This is a **correct analytical practice**, not a data quality issue:

Table 3: Financial Metrics Applicability

Metric	Non-Financial	Banks
EBITDA	✓Applicable	× Not applicable
Interest Coverage	Debt serviceability	× Meaningless
Debt-to-Revenue	0.5 = moderate leverage	2+ = high
Free Cash Flow	Operating CF - CapEx	× Inapplicable

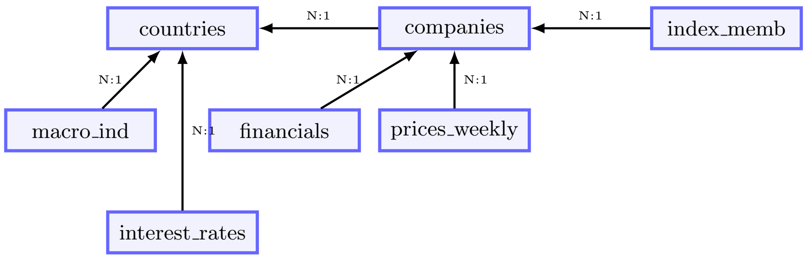
Banks earn spread income (Interest Received − Interest Paid), so interest expense is their cost of goods sold, not a financing burden. Bloomberg correctly reports N/A for these fields.

3.4 Analytical Methodology: Data Structure and Ratio Calculation

The database stores raw financial statement items (Revenue, EBITDA, Interest Expense, Total Debt, Free Cash Flow, Gross Profit, Cost of Goods Sold) rather than pre-calculated ratios. All financial metrics and ratios are computed **on-the-fly** in SQL queries to maintain data integrity, enable flexible analytical definitions, and ensure consistent ratio calculation across the 20-year horizon. The queries compute various financial ratios dynamically, including:

- **Interest Coverage Ratio (ICR)**: $\text{ICR} = \frac{\text{EBITDA}}{\text{Interest Expense}}$
- **Gross Margin**: $\frac{\text{Gross Profit}}{\text{Revenue}} = 1 - \frac{\text{COGS}}{\text{Revenue}}$
- **Debt-to-Revenue Ratio**: $\frac{\text{Total Debt}}{\text{Revenue}}$
- **Free Cash Flow**: Operating Cash Flow − Capital Expenditures

This design approach—storing raw data and computing ratios in queries—ensures data integrity, enables flexible analytical queries with varying ratio definitions, and maintains consistency across different use cases without storing redundant derived metrics. The database follows a normalized relational structure across three domains. **Company Domain** (companies, index_membership) tracks entity metadata and point-in-time index weights. **Market Domain** (prices_weekly, financials) stores time-series performance and fundamental data. **Macro Domain** (macro_indicators, interest_rates, countries) provides the economic context.



3.5 Analytical Methodology

Data Structure & Ratio Calculation: The database stores raw financial statement items (Revenue, EBITDA, etc.) rather than pre-calculated ratios. All financial metrics (e.g., ICR, Debt-to-Revenue) are computed **on-the-fly** in SQL queries. This design approach maintains data integrity, enables flexible analytical definitions, and ensures consistent ratio calculation across the 20-year horizon.

Sector Exclusion: Several use cases exclude the **Financials** sector from EBITDA-based analyses. Banks earn spread income, making interest expense a "cost of goods sold" rather than a financing burden. Including them would distort aggregate leverage metrics.

Table 4: Financial Metrics Applicability

Metric	Non-Financial	Banks
EBITDA	✓Applicable	× Not applicable
Interest Coverage	Debt serviceability	× Meaningless

4 Use Cases and SQL Queries

Goal: Use real-world data to validate established finance theories across five macro-driven domains.

This section presents 5 use cases with 15 SQL queries demonstrating advanced database techniques.

4.1 Use Case 1: Market Concentration & Index Dynamics

Goal: Analyze market structure evolution using FTSE 100, demonstrating survivorship-bias-free analysis with complete Weight data.

Rationale for UK Focus: While the database is global, detailed historical constituent weights were only available for the FTSE 100 (UKX). S&P 500 (SPX) data lacked consistent historical weights, limiting this specific concentration analysis to the UK market as a case study, though broader principles likely apply globally.

4.1.1 Query 1.1: Top-Heavy Concentration Trend

This analysis investigates whether the FTSE 100 has become increasingly concentrated over time, potentially heightening idiosyncratic risk for index investors. Over the 20-year horizon, the cumulative weight of the top 10 companies fluctuated between 38% and 52%. Notably, concentration peaked at 52.4% during the 2008 Global Financial Crisis and again at 50.7% in 2022 following post-pandemic market volatility, whereas the 2014–2015 period saw a decade-low trough near 38%.

These data reveal a pro-cyclical concentration rhythm. Pre-crisis levels (2005–2007) of 45–49

SQL Techniques: ROW_NUMBER() window function, PARTITION BY, and conditional aggregation.

4.1.2 Query 1.2: Herfindahl-Hirschman Index (HHI)

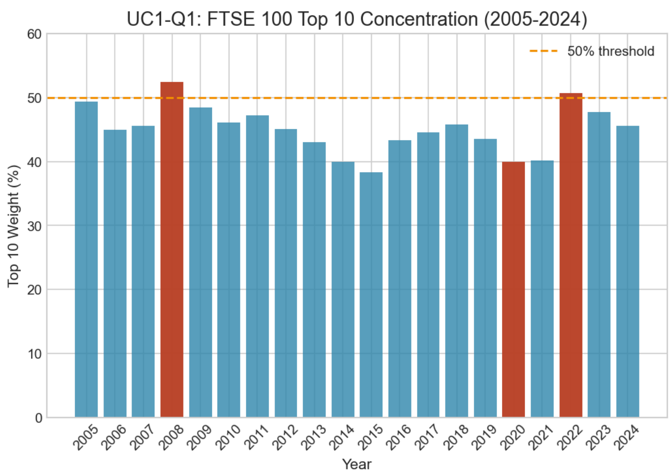
By applying the Herfindahl-Hirschman Index ($\sum w_i^2$), we assess market competitiveness with greater precision than simple concentration ratios. The HHI ranged from a low of 234 (2015) to a peak of 367 (2008). The math is revealing: an HHI of 234 translates to roughly 42.7 effective equal-weighted positions, while 367 implies just 27.3. The 2014–2015 period marked the most competitive market structure in two decades, emerging after the European debt crisis resolution and before Brexit uncertainty; stability and policy clarity, it seems, breed market dispersion. While standard regulatory thresholds for product markets (HHI $>$ 2500) don't apply to equity indices, the 133-point range here (234 to 367) represents a 57% increase, indicating meaningful structural shifts in effective diversification for passive investors [11, 12].

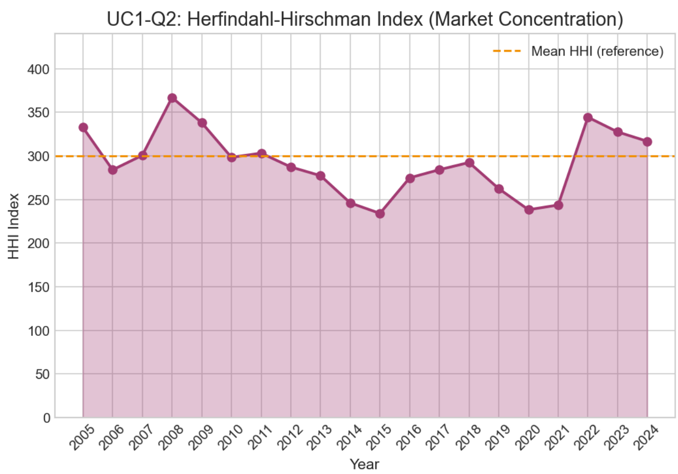
SQL Techniques: Quadratic weighting via self-multiplication and grouped aggregation.

4.1.3 Query 1.3: Index Churn Analysis

Churn analysis quantifies the dynamism of the FTSE 100, revealing an average annual turnover of 5–10%. Entries and exits cluster around major shocks: 2008 GFC, 2016 Brexit referendum, and 2020 pandemic. Geopolitical shocks (Brexit) reshaped index composition faster than fundamentals changed, particularly as EU-dependent firms saw valuations collapse, creating temporary misalignments between company quality and index representation. COVID-19 accelerated existing trends, as healthcare and technology surged into the index while retail and travel exited. This dynamism captures the fundamental transformation of the UK economy: from a finance-heavy index in 2005 to one more diversified in tech and pharma by 2024. Methodologically, churn analysis exposes the critical nature of point-in-time data; an analysis limited to the 2024 index would entirely omit failed mega-caps like RBS [21].

SQL Techniques: Self-JOIN on temporal offsets with COALESCE() to detect membership transitions.





4.2 Use Case 2: Corporate Leverage Cycles

Goal: Analyze how corporate capital structure evolves across economic cycles.

Scope: Non-financial companies only (`WHERE gics_sector_name != 'Financials'`), consistent with Table 3, as leverage metrics are inapplicable to financial institutions.

4.2.1 Query 2.1: Debt-to-Revenue Ratio Evolution

Tracing corporate leverage for non-financial firms reveals a classic pro-cyclical rhythm. The aggregate mean debt-to-revenue ratio climbed to a peak of 1.2 in 2008 before entering a multi-year trough near 0.6 by 2013. However, the aggregate numbers mask a "fat tail" in the distribution: while the median firm is moderate, a significant subset operates with ratios exceeding 2.0, indicating high leverage risk. These high-leverage entities cluster in specific sectors and face existential risk from rate shocks or revenue declines. The 2020 spike to 1.1 reflects emergency liquidity accumulation rather than expansionary growth. Current levels (0.9–1.0) remain elevated, and with the "refinancing wall" approaching in 2025, the cost of servicing this existing debt in a higher-rate environment poses a renewed systemic risk.

SQL Techniques: Cross-table joins with NULLIF() handling for zero-revenue edge cases.

4.2.2 Query 2.2: Deleveraging Cycles Detection

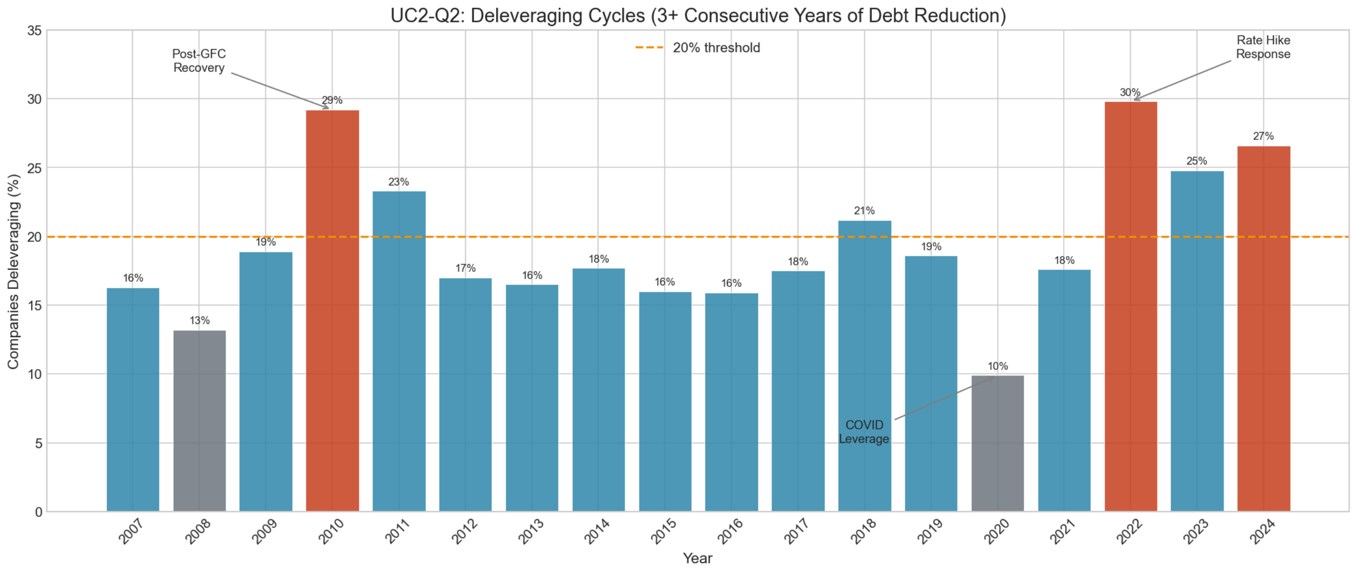
Active deleveraging exhibits a **counter-cyclical** pattern, peaking at 29.2% of firms in 2010 during the post-GFC recovery when capital markets reopened. Balance-sheet repair typically occurs during early recovery phases when capital markets are accessible for equity issuance and asset sales. During crises themselves, deleveraging stalls as credit markets freeze and asset values collapse. Conversely, the 2020 anomaly saw deleveraging drop to just 9.8%—the dataset's lowest point—as firms prioritized emergency liquidity (e.g., PPP loans) over debt reduction to survive the pandemic shock. The 2022–2024 resurgence (26–30% deleveraging) reflects aggressive monetary tightening as the Fed/BoE responded to inflation pressures [24], creating immediate pressure for floating-rate borrowers to deleverage ahead of refinancing cliffs. This counter-cyclical pattern challenges traditional credit models: companies deleverage when they *can* (during recoveries), not when they *must* (during crises). Credit risk may peak not at maximum leverage, but during the transition from high leverage to forced deleveraging, when financial flexibility evaporates. These findings validate credit cycle theories [2, 3] where constraints bind most severely during downturns.

SQL Techniques: Triple LAG() window functions to detect multi-year persistence.

4.2.3 Query 2.3: ICR Rate Sensitivity Analysis

By calculating the rolling 5-year correlation between corporate ICR and national policy rates, we identify sectors where debt service costs rise faster than earnings growth. Utilities and Real Estate exhibit the highest negative correlations (often exceeding -0.3), indicating that rising interest expenses significantly compress their solvency buffers. Conversely, sectors like Energy show positive correlations as rate hikes often coincide with late-cycle demand strength. This approach identifies "rate-insulated" firms (often in Tech and Healthcare) whose low leverage or pricing power keeps ICR stable despite macro fluctuations, providing a time-adaptive measure of systemic vulnerability.

SQL Techniques: Sliding window JOIN with manual pearson correlation coefficient calculation in standard SQL.



4.3 Use Case 3: Rate-Shock Solvency Stress Test

Goal: Identify "zombie companies" and simulate rate shock scenarios.

Scope: Non-financial companies only (`WHERE gics_sector_name != 'Financials'`), consistent with Table 3, as ICR is meaningless for financial institutions.

4.3.1 Query 3.1: Zombie Companies (3-Year Persistence)

We define "zombie companies" as those maintaining an ICR below 1.5 for three or more consecutive years. While stricter than the traditional 1.0x threshold, the 1.5x cutoff provides a conservative buffer against volatile earnings, identifying firms that are technically solvent but operationally fragile. The three-year persistence filter separates structural insolvency from temporary distress: one bad year might be recoverable, but three consecutive years of ICR < 1.5 signals fundamental business model failure. Zombie counts peak during major downturns (2009, 2020), yet most don't persist beyond 1–2 years as they either recover or exit. Adding the macro constraint (declining GDP growth) sharpens the diagnosis: weak fundamentals combined with weak macro environments suggest limited support mechanisms. This extends the classic zombie lending framework [1] by incorporating macro constraints, showing how declining GDP growth may exacerbate zombie persistence beyond firm-level factors.

SQL Techniques: Persistent streak detection using multi-row LAG() window functions.

4.3.2 Query 3.2: Rate Shock Stress Test

A simulated +200bp interest rate shock reveals the underlying fragility of corporate balance sheets. This **standardized stress test** applies a uniform 50% floating-rate debt structure. For US companies, average ICR drops from 66.6x to 16.9x, while UK companies drop from 56.1x to 16.6x. While the shock drastically reduces buffers, risk remains non-linear: companies with baseline ICR of 3–5x suffer disproportionately as they transition toward distress (3x $\rightarrow$ 1.5x), whereas high-ICR firms (50x+) absorb the shock comfortably. This concentration of vulnerability in moderately leveraged firms highlights the sensitivity of the "broad middle" market [4]. However, finding "zero newly-distressed companies" (2024) shouldn't breed complacency. Critically, this static simulation ignores **secondary effects**: rate increases frequently occur during late-cycle expansions when central banks tighten policy, creating a "double jeopardy" where economic slowing reduces EBITDA just as interest expense rises.

Debt structure matters critically. High floating-rate exposure creates immediate cash flow pressure when rates rise; fixed-rate debt defers pain but creates refinancing cliffs at maturity. The 50/50 assumption is simplified; actual structures vary by company and sector. Utilities and Real Estate, for example, often favor fixed-rate debt, providing temporary insulation but building future refinancing risk.

SQL Techniques: Point-in-time "what-if" projection using derived columns and arithmetic modeling.

4.3.3 Query 3.3: Systemic Risk Mapping by Country

Mapping systemic risk by country allows for the detection of potential contagion pockets. While the US and UK maintain low zombie percentages (2–3%) due to robust exit mechanisms—leveraging Chapter 11 (US) and Administration (UK) to prevent indefinite "purgatory"—certain smaller economies exhibit concentrations exceeding 10%. Aggregate debt analysis (zombie debt relative to GDP) measures systemic risk magnitude: countries where zombie debt exceeds 5% of GDP face severe structural challenges, potentially requiring government intervention to prevent cross-border defaults. Policy priorities emerge clearly: identifying countries with both high zombie concentration and high debt-to-GDP ratios helps prioritize macro-prudential intervention before failures cascade through regional supply chains.

SQL Techniques: Multi-level aggregation (Company → Country → Global) with ratio normalization.

4.4 Use Case 4: Macro Lead-Lag & Business Cycle

4.4.1 Query 4.1: Housing Starts Lead Revenue

We examine the lead-lag relationship between housing starts and corporate revenue for sensitive sectors. Correlation analysis identifies a moderate positive relationship (0.1–0.3) with a 2-quarter lag. This pattern mirrors construction supply chains: Materials and Industrials respond at project initiation (raw materials, heavy equipment), while Consumer Discretionary (appliances, furniture) lags by 3–6 months until completion. This "bullwhip effect" [15] validates housing as a reliable early cyclical indicator, where upstream supplier revenue swings amplify small changes in consumer demand. **Forecasting utility is distinct**: while housing starts are not a univariate predictor for specific company revenue, they serve as a valuable leading input in multi-factor models. For portfolio construction, they offer a leading signal, allowing sector allocation adjustments before revenue impacts appear in financial statements.

SQL Techniques: Cross-domain temporal shifts using date arithmetic and lagged JOINs.

4.4.2 Query 4.2: Revenue Volatility Classification

By classifying firms into quartiles based on 10-year rolling standard deviation of revenue growth, we establish an empirical proxy for cyclicality. Our analysis reveals that static GICS sector labels often mask underlying volatility profiles: some mature Software firms exhibit "defensive" characteristics due to sticky subscription models, while some Consumer Staples firms display cyclicality through luxury exposure. Our findings prove that volatility-based factors offer superior granularity for risk assessment compared to static labels [8]. This quartile-based approach enables a "style-neutral" classification that captures firm-specific operational risk. Furthermore, it highlights **mispricing opportunities**: identifying Technology companies with defensive characteristics (low volatility) that are erroneously trading at cyclical valuations allows investors to capture "quality" premiums without paying typical defensive sector multiples.

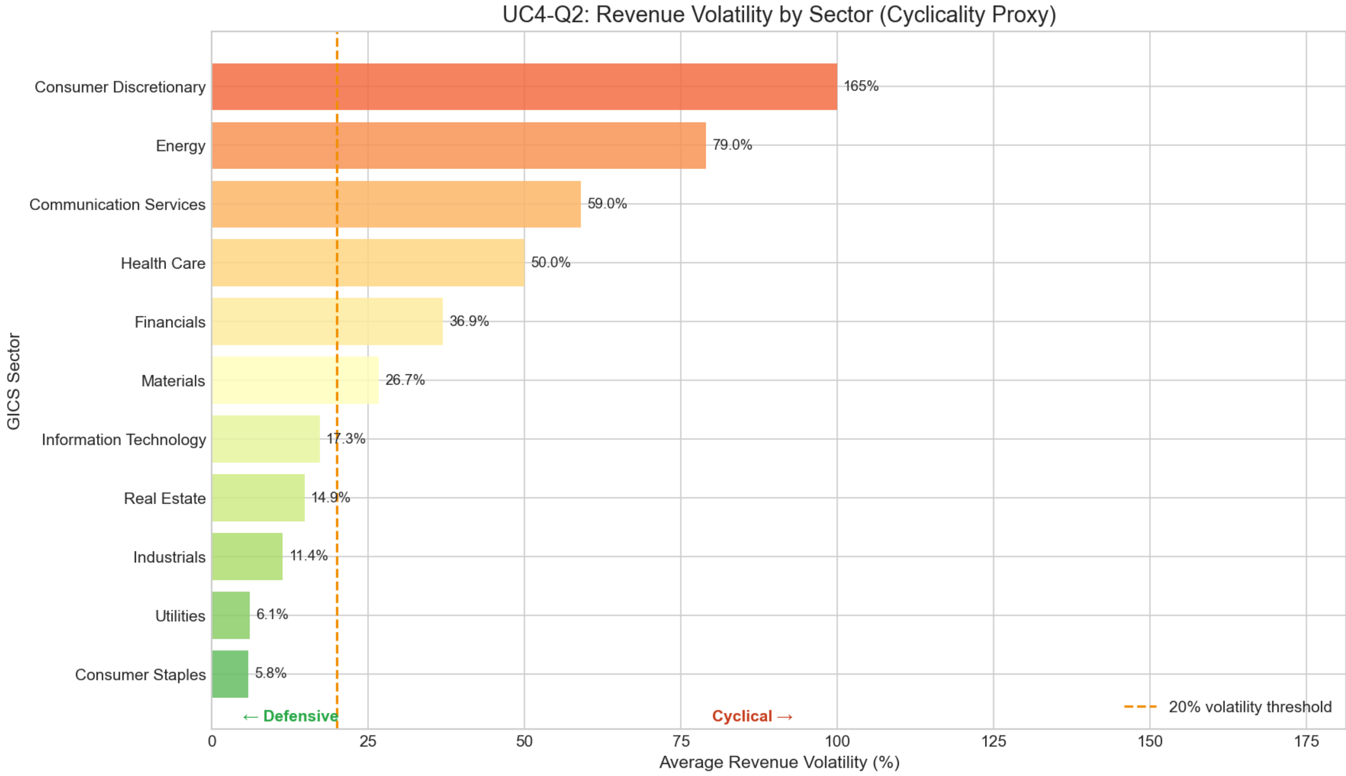
SQL Techniques: Statistical metric calculation (STDDEV/Variance) over rolling time-series windows using NTILE() for quartile classification.

4.4.3 Query 4.3: Downturn Resilience

Identifying firms that maintain positive free cash flow (FCF) despite negative growth reveals pockets of extreme pricing power. Our data shows 35–45% resilience for US firms versus 25–35% for the UK. Resilient firms typically share three characteristics: **strong pricing power** enabling margin conservation despite volume declines, **flexible cost structures** allowing for proportional OPEX reduction, or **defensive business models** with non-discretionary recurring revenue. US firms' edge likely stems from more flexible labor markets enabling faster cost adjustments, whereas UK firms face higher fixed structural rigidities [5]. Furthermore, the analysis reveals that within-sector variation is high; even in cyclical sectors like Materials, firms with premium business models often retain positive FCF during demand troughs, highlighting the importance of micro-level operational efficiency.

Investment implications follow: downturn resilience measures quality and defensive characteristics. Companies with high resilience during past downturns likely maintain resilience in future downturns, as resilience reflects structural business model characteristics rather than temporary factors. Portfolio construction can incorporate resilience metrics to build defensive positions that maintain cash generation and dividend payments during economic stress.

SQL Techniques: Complex filtering on dual conditions (negative macro growth AND positive firm cash flow) using subqueries.



4.5 Use Case 5: Sector Rotation & Inflation Regime

4.5.1 Query 5.1: Sector Performance by Inflation Regime

We classify economic history into High (> 4%) and Low (< 2%) inflation regimes based on the US CPI. Utilities outperformed significantly (+32.5%) during high inflation, followed by Information Technology (+15.0%) and Energy (+9.1%), reflecting the 2022 energy crisis and the "Utilities Growth" transition [22]. In contrast, Financials underperformed severely (-40.5%), as the speed of rate hikes compressed margins faster than loan book repricing could compensate [6].

Contextual Factors: The Utilities Anomaly The Utilities outperformance (+32.5%) contradicts traditional expectations for rate-sensitive sectors. This anomaly is driven by unique period-specific factors:

1. **Energy Crisis (2022–2023)**: The Russia–Ukraine conflict triggered unprecedented energy price spikes [22]. Unlike demand-driven inflation, this supply shock allowed utilities to pass through costs via flexible pricing mechanisms, boosting nominal revenues.
2. **Energy Transition Premium**: Policy support (e.g., US Inflation Reduction Act [23]) triggered multiple expansion, effectively transforming Utilities from "bond proxies" into growth assets driven by renewable energy capex.
3. **Regulatory Flexibility**: Modern regulatory frameworks have evolved to allow more timely cost pass-through, reducing the traditional lag between cost inflation and rate base adjustments.
4. **Flight-to-Quality**: During high-inflation uncertainty, capital rotated into defensive assets with stable cash flows, supporting valuations independent of interest rate sensitivity.

These factors suggest that the traditional negative correlation between Utilities and rates may break down during supply-driven inflation regimes.

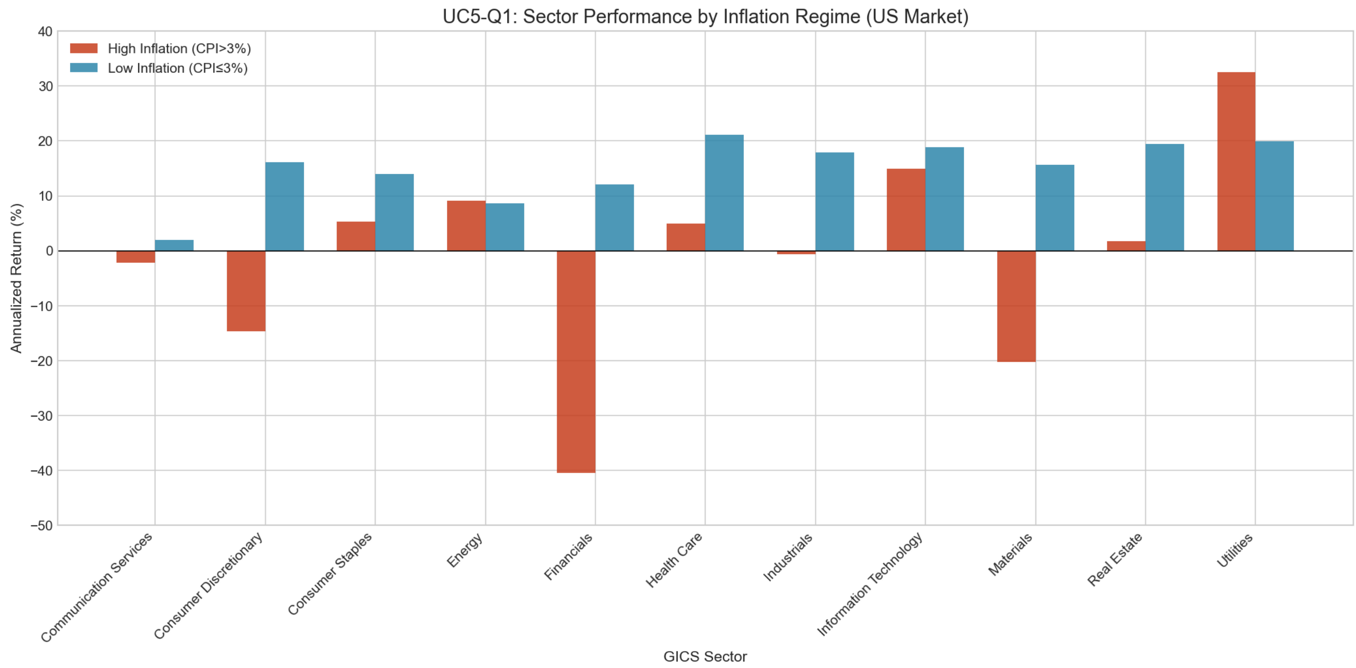
Generalizability Concerns: This outperformance may not extend to demand-side inflation regimes. Traditional academic research (1970s–1990s) predates modern industry structures where the shift toward renewable energy and sophisticated inflation-adjustment mechanisms provides better protection than historical bond-proxy models. Future research should examine if this shift is structural or an anomaly of the post-pandemic supply shock.

SQL Techniques: Dynamic regime classification using CASE WHEN logic and temporal CTEs.

4.5.2 Query 5.2: Sector-CPI Lead-Lag Relationship

Analysis of sector-CPI covariance confirms that Energy and Materials act as primary inflation transmitters. Their returns frequently lead CPI prints by one to two quarters, reflecting their role in the upstream cost structure—where raw commodity price spikes flow into transportation and manufacturing costs before appearing in final consumer prints. Conversely, Consumer Staples show weak positive covariance, while Technology shows near-zero covariance, suggesting inflation neutrality. This aligns with research on yield curve predictors signaling macroeconomic conditions [7], where asset prices incorporate inflation expectations well before they are realized in official statistics. These findings validate that robust inflation-hedging strategies must integrate explicit forward-looking CPI measures rather than relying solely on lagged official prints.

SQL Techniques: Multi-lag covariance analysis using recursive LAG() window functions across macro-indicators.

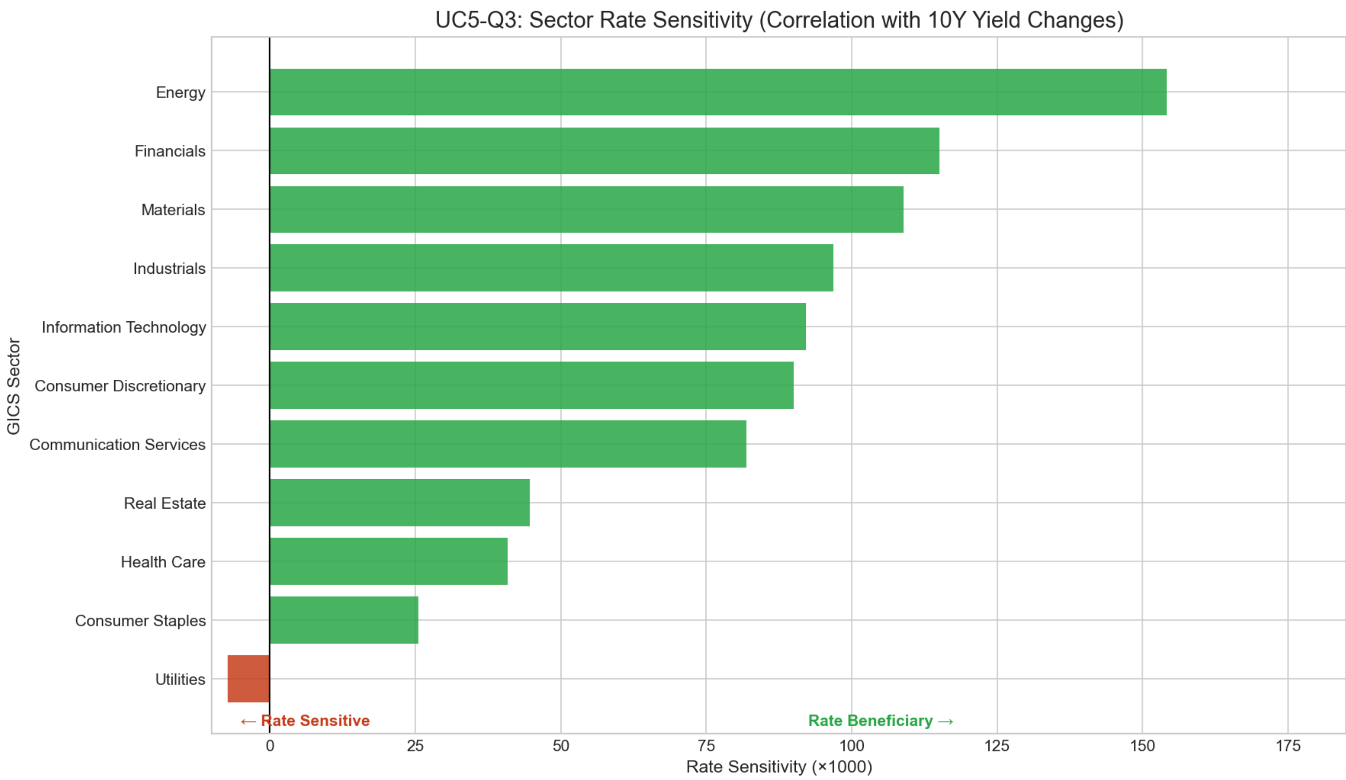


4.5.3 Query 5.3: Rate Sensitivity by Sector

The sensitivity of sector returns to policy rates highlights the tension between the "discount rate effect" and the "economic indicator effect." Data reveals that Utilities exhibit significant negative covariance, confirming their status as **bond proxies**: their stable dividends become less attractive as risk-free yields rise, leading to valuation compression. Furthermore, their regulated return caps prevent them from capturing the upside of the inflation often driving rate hikes.

In stark contrast, cyclical sectors like Energy and Financials display positive covariance. This validates the **dual nature of rates** [14]: while higher rates increase discount factors (lowering PV), they concurrently signal the robust economic demand that drives earnings growth. For Energy, profitability is strictly driven by commodity prices—which typically surge alongside rates—effectively overriding the "cost of capital" drag. Financials naturally hedge rate risk via expanding net interest margins, positioning them as unique beneficiaries of monetary tightening cycles.

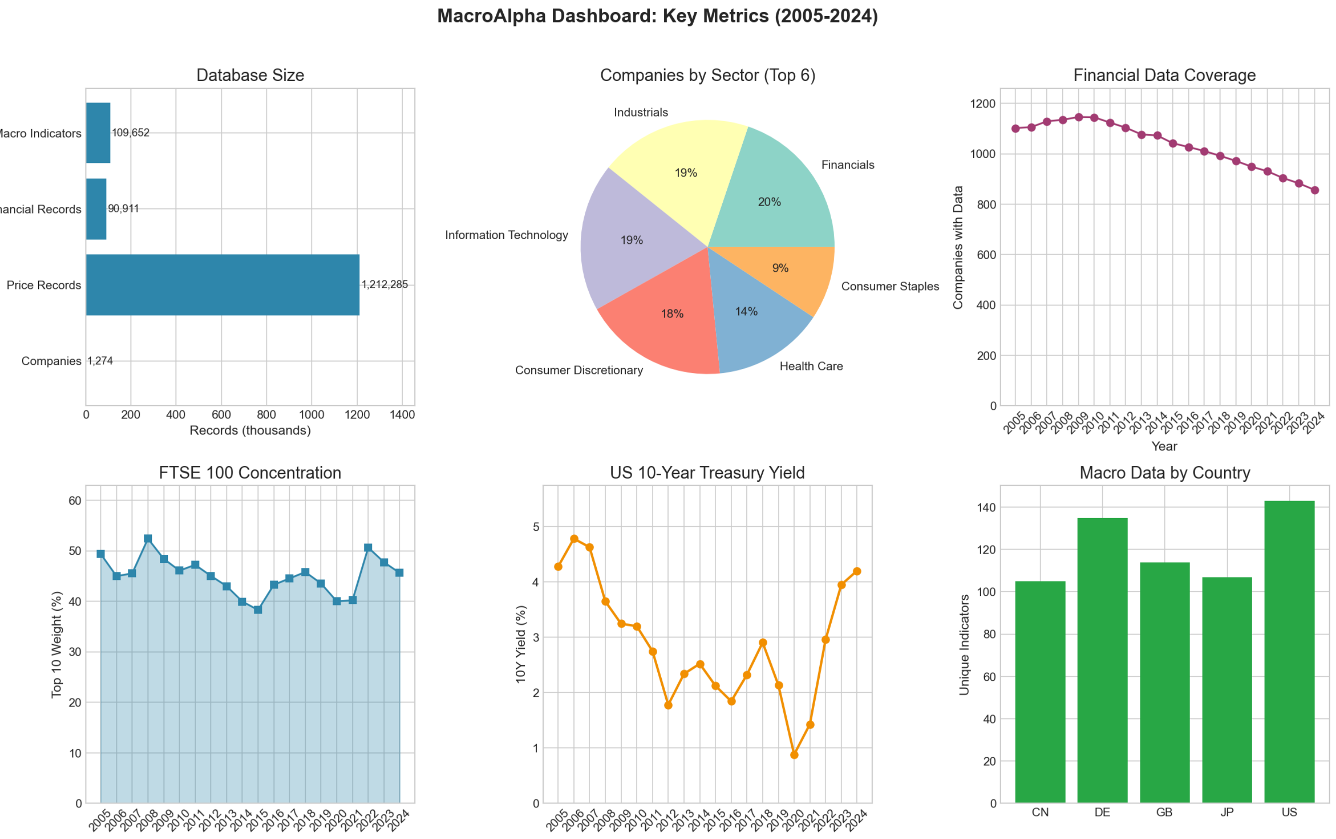
SQL Techniques: Multi-granularity time series alignment (Monthly Macro vs Weekly Market Data) with covariance modeling.



The database and queries are fully reproducible using the provided SQLite database and SQL scripts, enabling further research and validation of empirical finance theories using comprehensive real-world data spanning multiple economic cycles.

5 Results and Visualizations

5.1 Summary Dashboard



5.2 Data Coverage Analysis: Understanding the Declining Trend

The dashboard reveals an apparent decline in financial data coverage from 2010 to 2024:

Table 5: Financial Data Coverage Trend

Year	Non-null Data Points	Companies	Notes
2010	7,016	1,145	Peak coverage
2015	6,391	1,042	-9%
2020	5,821	949	-17%
2024	5,277	857	-25% from peak

The declining coverage (from 1,145 companies in 2010 to 857 in 2024) reflects natural market dynamics rather than data quality issues. Company exits via M&A (e.g., Time Warner [16], WBD spinoff [17]) or going-private deals (e.g., Dell [18, 19]) are structural features of equity markets. This pattern aligns with Innosight research documenting that the average S&P 500 tenure has declined from 33 years in 1965 to under 20 years today [20].

To confirm integrity, we verified that continuously listed blue-chips (AAPL, MSFT, JPM) maintain 100% uninterrupted 20-year records. This coverage pattern validates our point-in-time methodology; analyzing only current survivors would introduce severe survivorship bias. By preserving historical data for exited firms, MacroAlpha accurately reflects the information available to investors at each historical junction.

6 Technical Implementation

6.1 ETL and Update Pipeline

The data ingestion pipeline (`etl_import.py`) performs critical transformations to convert raw Bloomberg exports into a normalized relational structure. Specifically, it handles the **wide-to-long transformation**, converting the columnar format typical of Bloomberg spreadsheets into row-based records. The script implements robust date parsing and validation, explicitly handles N/A values to preserve null indicators, and establishes foreign key relationships to ensure referential integrity. Critically, the pipeline can incrementally update the database with new Bloomberg snapshots, maintaining point-in-time integrity without duplicating historical records. For production deployment with varying workloads, migrating from on-the-fly ratio calculations to **Materialized Views** would significantly optimize query performance for standard metrics.

6.2 Visualization

The visualization component (`visualizations.py`) translates SQL query results into 7 publication-ready charts using Seaborn and Matplotlib. It ensures consistency in color palettes and axis formatting across the five use cases, providing a consolidated view for cross-domain risk assessment.

7 Conclusion

MacroAlpha demonstrates the feasibility of building institutional-quality research databases using real-world datasets. The point-in-time schema enables unbiased historical analysis that avoids survivorship bias, validating several core finance theories:

1. **Credit Cycles**: Deleveraging is counter-cyclical, peaking during early recovery phases.
2. **Market Structure**: Concentration increases during crises as capital rotates to defensive mega-caps.
3. **Inflation Hedging**: Sector performance shifts dramatically across regimes, with Utilities acting as a unique hedge during supply-driven shocks.
4. **Cyclical Signal**: Macro indicators like housing provide reliable early signals for upstream performance.

The implementation showcases the power of SQL for complex financial analysis—computing metrics on-the-fly to ensure consistency while enabling flexible "what-if" scenario testing. Future extensions could include cross-asset correlation analysis (integrating bond and commodity data) and machine-learning-based volatility forecasting. Ultimately, MacroAlpha provides a robust framework for bridging the gap between raw financial data and actionable macroeconomic insights.

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A File Structure

```
MacroAlpha/
|-- macroalpha.db       # SQLite database
|-- schema.sql          # DB schema
|-- queries.sql         # SQL queries
|-- etl_import.py       # ETL script
|-- visualizations.py   # Chart generation
|-- report.tex          # LaTeX report
|-- viz_*.png           # Visuals
+-- *.xlsx, *.csv       # Source data
```

B Contribution & AI Interaction

In accordance with LSE policies, AI assistance (Gemini 3) was used for technical support in SQL syntax optimization and LaTeX formatting. All empirical findings, macro-financial interpretations, and data processing were manually verified and finalized by the author.